

SANGJU, Korea, Republic of

WMO: 471370

Lat: 36.4083N

Lon: 128.1575E

Elev: 320

StdP: 14.53

Time Zone: 9.00 (E09)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	17.7	-7.4	3.7	21.1	-3.9	4.5	23.4	20.8	27.8	17.7	29.7	6.3	270	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	91.7	76.6	89.2	75.5	86.7	73.8	79.0	87.2	77.8	85.5	76.6	83.8	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	141.0	83.2	75.6	135.4	82.1	74.5	130.6	81.0	42.9	87.6	41.6	85.6	40.4	84.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	13.0	11.0	9.8	94.8	4.4	2.2	6.6	96.4	4.0	97.8	1.6	99.0	-1.6	100.6
			7.0	80.5	3.9	1.5	4.2	81.6	1.9	82.4	-0.4	83.3	-3.2	84.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	30.6	34.9	44.4	55.3	65.5	72.6	77.5	78.1	69.2	58.1	45.6	33.5	(6)
(7)		DBStd	17.66	6.23	6.53	6.94	6.37	4.98	4.15	4.77	5.04	4.88	5.71	7.02	6.50	(7)
(8)		HDD50	1937	603	423	201	23	0	0	0	0	0	8	168	511	(8)
(9)		HDD65	4699	1068	843	639	296	54	3	0	16	223	581	976	(9)	
(10)		CDD50	3963	0	0	27	182	480	679	852	872	575	259	37	0	(10)
(11)		CDD65	1252	0	0	0	5	70	232	387	407	141	10	0	0	(11)
(12)		CDH74	10000	0	0	2	106	682	1694	3219	3557	704	36	0	0	(12)
(13)		CDH80	3186	0	0	0	15	132	468	1118	1336	116	1	0	0	(13)
(14)	Wind	WSAvg	4.0	5.5	5.1	5.6	5.0	3.9	3.0	2.6	2.5	2.3	2.7	4.0	5.4	(14)
(15)	Precipitation	PrecAvg	47.2	0.8	1.2	2.0	3.5	3.6	4.8	11.2	10.3	5.4	1.9	1.6	1.0	(15)
(16)		PrecMax	72.9	2.9	2.8	3.6	7.3	7.6	10.3	25.3	20.7	14.9	5.8	4.6	1.7	(16)
(17)		PrecMin	30.7	0.1	0.1	0.3	1.1	0.9	1.1	4.0	2.5	1.3	0.1	0.2	0.3	(17)
(18)		PrecStd	10.1	0.7	0.7	0.9	1.9	1.8	2.7	5.7	5.5	3.7	1.5	1.2	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	57.8	71.2	82.2	87.3	90.5	94.5	95.8	86.7	77.7	69.3	53.5	(19)
(20)		0.4%	MCWB	41.0	46.0	53.4	60.0	62.7	68.6	77.6	77.0	74.8	64.3	58.4	45.8	(20)
(21)		2%	DB	46.0	53.1	65.4	76.9	83.5	87.3	91.3	92.5	83.0	74.3	64.2	49.6	(21)
(22)		2%	MCWB	37.8	42.7	49.5	56.5	62.0	68.2	77.1	77.3	71.0	62.7	54.4	42.3	(22)
(23)		5%	DB	42.7	49.4	61.2	73.2	80.7	85.1	88.8	90.0	80.7	71.4	60.6	46.5	(23)
(24)		5%	MCWB	35.6	40.0	46.8	54.5	61.3	67.9	76.5	77.0	69.3	60.9	52.8	40.2	(24)
(25)		10%	DB	40.0	45.9	57.3	69.3	78.0	82.6	86.2	87.4	78.3	68.5	57.2	43.8	(25)
(26)		10%	MCWB	33.5	37.6	45.1	52.6	60.5	67.4	75.4	76.3	68.3	59.2	50.0	38.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	50.9	56.0	64.5	70.3	76.1	80.1	80.8	77.1	69.0	60.6	48.2	(27)
(28)		0.4%	MCD B	47.4	53.6	66.1	77.3	77.1	82.7	88.3	90.4	84.1	73.0	66.6	52.4	(28)
(29)		2%	WB	39.1	43.9	52.5	60.0	66.5	73.4	78.9	79.3	74.6	66.0	56.8	43.8	(29)
(30)		2%	MCD B	43.7	50.7	60.4	71.3	76.8	80.4	86.7	88.1	79.4	71.4	62.1	47.5	(30)
(31)		5%	WB	36.5	40.7	49.4	57.0	64.5	71.7	77.7	78.2	72.6	63.2	54.0	41.0	(31)
(32)		5%	MCD B	41.0	47.0	58.9	67.8	75.1	79.2	85.4	86.3	76.8	68.6	59.3	45.5	(32)
(33)		10%	WB	34.3	38.4	46.5	54.5	62.7	70.1	76.6	77.1	70.5	60.5	51.1	38.5	(33)
(34)		10%	MCD B	39.1	44.8	55.6	65.0	72.9	77.9	84.0	84.5	75.6	67.1	56.3	43.5	(34)
(35)	Mean Daily Temperature Range		MDBR	14.5	16.6	19.0	21.2	20.9	17.5	12.7	13.7	16.0	19.9	17.2	14.6	(35)
(36)		5% DB	MCD BR	17.6	22.5	25.9	29.2	26.7	22.3	16.6	18.0	19.1	22.3	21.7	18.1	(36)
(37)		5% DB	MCWBR	11.9	14.4	13.9	12.7	10.6	7.7	5.8	6.3	8.4	11.3	12.9	12.7	(37)
(38)		5% WB	MCD BR	16.1	20.4	22.6	23.5	20.4	15.8	14.4	15.4	14.1	17.7	18.1	16.8	(38)
(39)		5% WB	MCWBR	11.9	14.5	14.1	12.2	10.2	7.1	6.0	6.6	7.5	10.2	12.2	12.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.458	0.542	0.575	0.585	0.644	0.600	0.551	0.479	0.457	0.435	0.403	(40)	
(41)		taud	2.014	1.927	1.764	1.741	1.763	1.701	1.867	1.988	2.132	2.062	2.041	2.050	(41)	
(42)		Ebn at Noon	235	239	229	231	232	218	226	235	246	237	225	227	(42)	
(43)		Edh at Noon	42	52	67	72	71	75	64	55	45	44	41	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	824	1057	1372	1573	1774	1623	1368	1383	1249	1135	824	736	(44)	
(45)		RadStd	47	96	134	122	158	126	182	188	142	95	101	53	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+272	+212	(46)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+76	(47)

Nomenclature: See separate page